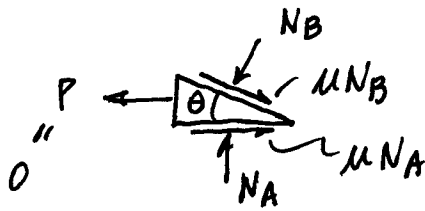


Self-locking Wedge (same μ for top & bottom of wedge)



- Pull out case.
- Set $P = 0$

$$+\uparrow \sum F_y = 0; \quad N_A - N_B \cos \theta - \mu N_B \sin \theta = 0$$

$$\textcircled{1} \quad N_A = N_B (\cos \theta + \mu \sin \theta)$$

$$+\rightarrow \sum F_x = 0; \quad \mu N_A + \mu N_B \cos \theta - N_B \sin \theta = 0$$

$$\textcircled{2} \quad N_A = \frac{1}{\mu} N_B [\sin \theta - \mu \cos \theta]$$

Set $\textcircled{1} = \textcircled{2}$:

$$\cancel{N_B} (\cos \theta + \mu \sin \theta) = \frac{1}{\mu} \cancel{N_B} [\sin \theta - \mu \cos \theta]$$

Mult
by μ

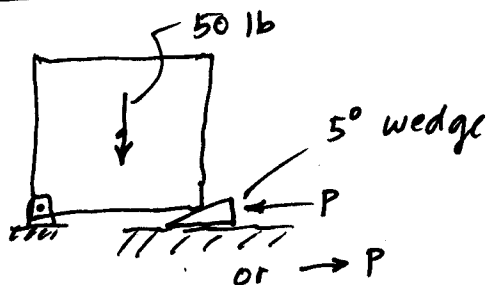
$$\mu \cos \theta + \mu^2 \sin \theta = \sin \theta - \mu \cos \theta$$

$$2\mu (\cos \theta) = \sin \theta (1 - \mu^2)$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2\mu}{(1 - \mu^2)}$$

Self-locking
Condition

$$\tan \theta = \frac{2\mu}{(1 - \mu^2)}$$



$\mu = .2$ at top
& bottom

Is this wedge self-locking?

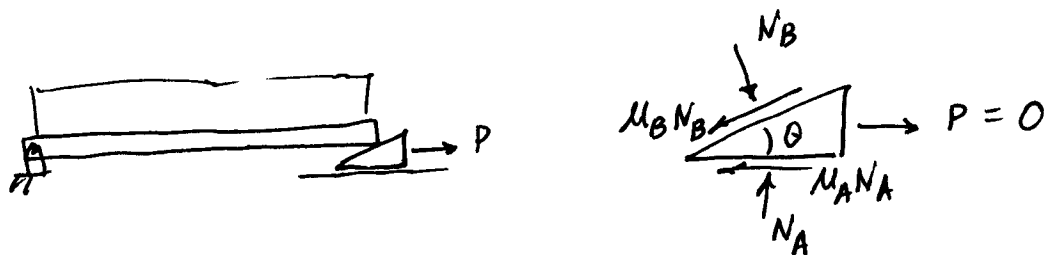
$$\theta = \tan^{-1} \left(\frac{2(.2)}{1 - (.2)^2} \right)$$

$$= \tan^{-1} (.4167)$$

$$\theta = 22.6^\circ$$

At angles $< 22.6^\circ$, this wedge
is self-locking.

Wedge Self-locking Equation, with $2 \mu's$



$$+ \uparrow \quad N_A - N_B \cos \theta - \mu_B N_B \sin \theta = 0$$

$$\textcircled{1} \quad N_A = N_B (\cos \theta + \mu_B \sin \theta)$$

$$\leftarrow + \quad \mu_A N_A + \mu_B N_B \cos \theta - N_B \sin \theta = 0$$

$$\textcircled{2} \quad N_A = \frac{1}{\mu_A} N_B (\sin \theta - \mu_B \cos \theta)$$

$$\textcircled{1} = \textcircled{2} \quad \cancel{N_B} () = \frac{1}{\mu_A} \cancel{N_B} ()$$

N_B
cancels

$$\mu_A (\cos \theta + \mu_B \sin \theta) = (\sin \theta - \mu_B \cos \theta)$$

$$(\cos \theta) (\mu_A + \mu_B) = \sin \theta (1 - \mu_A \mu_B)$$

$$\tan \theta = \frac{\mu_A + \mu_B}{1 - \mu_A \mu_B}$$

$$\theta = \tan^{-1} \left(\frac{\mu_A + \mu_B}{1 - \mu_A \mu_B} \right)$$

Compare this to the case where $\mu_A = \mu_B = \mu$:

$$\theta = \tan^{-1} \left(\frac{2\mu}{1 - \mu^2} \right)$$